Determinant of 2x2 matrices

Basic skills

Q1. a) $\begin{vmatrix} 5 & 3 \\ 1 & 2 \end{vmatrix} = 5 \times 2 - 3 \times 1 = 10 - 3 = 7$

trail

lead

b) $\begin{vmatrix} -3 & -1 \\ 2 & 1 \end{vmatrix} = -3 \times 1 - -1 \times 2 = -3 + 2 = -1$

c) $\begin{vmatrix} 4 & 2 \\ -2 & 4 \end{vmatrix} = 4 \times 4 - 2 \times -2 = 16 + 4 = 20$

Q2. As area SF is $|M|$ its $8 \times |M|$

$|M| = \begin{vmatrix} 3 & 1 \\ 4 & 2 \end{vmatrix} = 6 - 4 = 2$ so new area $8 \times 2 = 16$

Q3. As reflection preserves area so determinant must be 1.

Q4. To preserve area $|M| = 1$

$\Rightarrow \begin{vmatrix} a & 0 \\ 0 & d \end{vmatrix} = 1 \Rightarrow ad - 0 = 1 \Rightarrow a = \frac{1}{d}$

Competency skills

Q5. $|M| = 5$

$$\Rightarrow \begin{vmatrix} x & -3 \\ 3 & x-5 \end{vmatrix} = 5 \Rightarrow x(x-5)+9=5$$

$$\Rightarrow x^2 - 5x + 4 = 0 \Rightarrow (x-4)(x-1)=0$$

so $x = 1$ or 4

Q6.

$$\begin{vmatrix} x & y \\ 5 & 1 \end{vmatrix} = 2 \Rightarrow x - 5y = 2$$

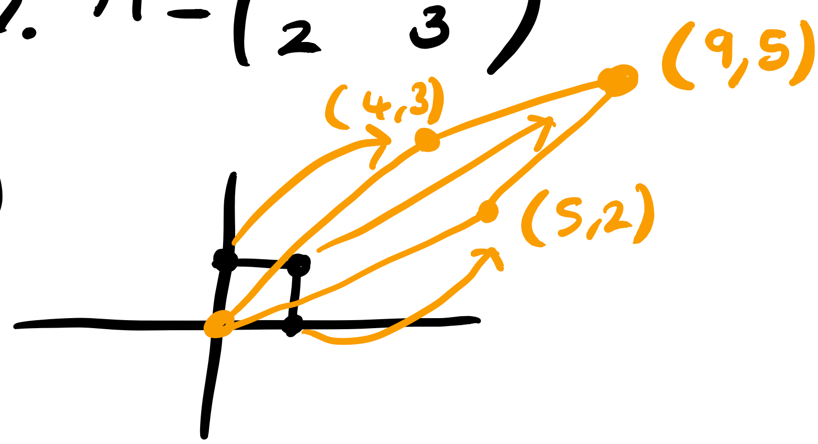
$$\begin{vmatrix} x & 2 \\ y & 3 \end{vmatrix} = 5 \Rightarrow 3x - 2y = 5$$

so
$$\begin{array}{r} 3x - 15y = 6 \\ - \quad 3x - 2y = 5 \end{array} \Rightarrow -13y = 1 \quad y = -\frac{1}{13}$$

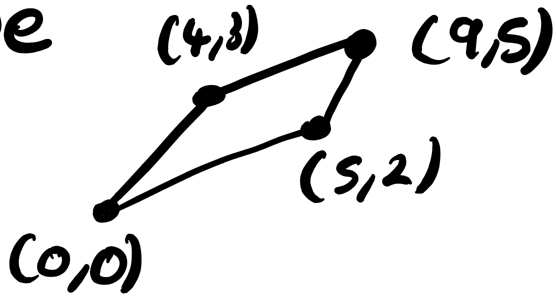
and $x = 2 + 5y = 2 - \frac{5}{13}$ so $x = \frac{21}{13}$

Q7. $M = \begin{pmatrix} 5 & 4 \\ 2 & 3 \end{pmatrix}$

a)



b) $|M| = 15 - 8 = 7$ so new area of shape



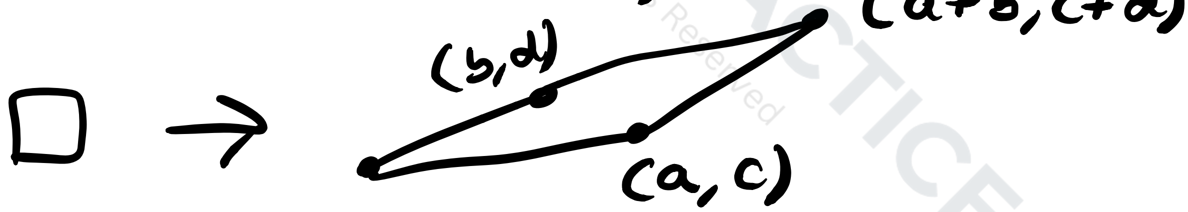
is 7

Q8. Let $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \Rightarrow kM = \begin{pmatrix} ka & kb \\ kc & kd \end{pmatrix}$

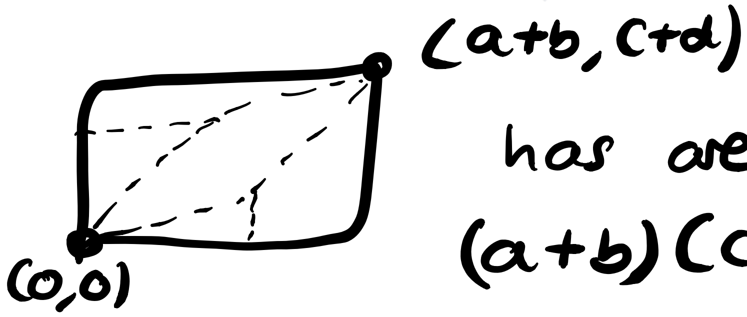
so $|kM| = \begin{vmatrix} ka & kb \\ kc & kd \end{vmatrix} = ka \times kd - kb \times kc$
$= k^2ad - k^2bc$

$= k^2(ad - bc) = k^2|M|$

Q9. Under $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

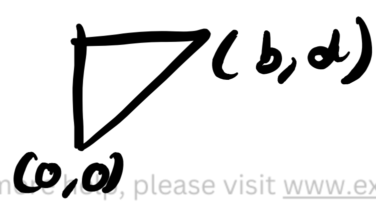


now full rect



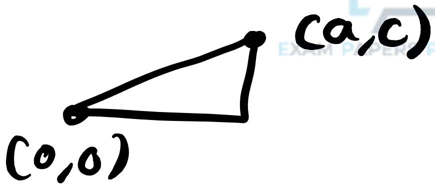
has area $(a+b)(c+d)$

now remove



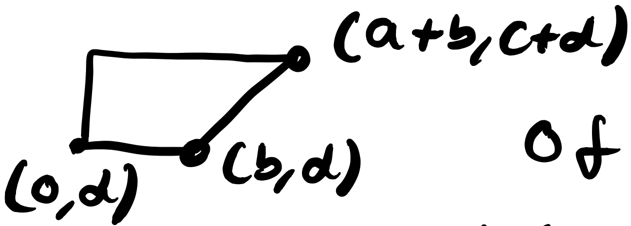
of area $\frac{1}{2}bd$

and remove



of area $\frac{1}{2}ac$

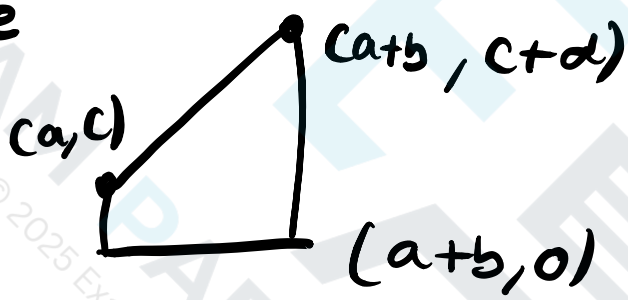
and remove



of area

$$\frac{1}{2}(b + a+b) \times c$$
$$= (b + \tfrac{1}{2}a)c$$

and remove



of area

$$\frac{1}{2}(c + c+d) \times b$$
$$= (c + \tfrac{1}{2}d)b$$

so $(a+b)(c+d) - \frac{1}{2}bd - \frac{1}{2}ac$

$- (b + \frac{1}{2}a)c \quad - (c + \frac{1}{2}d)b$

$= \cancel{ac} + bc + ad + \cancel{bd} - \cancel{bd} - \cancel{ac} - 2bc$

$= ad - bc$

as required

as $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$

Q10. $y = x+1$ mapped to $(1,2)$

a) so for any x: $\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} x \\ x+2 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$

$\Rightarrow$ $xa + xb + 2b = 1$

$xc + xd + 2d = 2$

so let $a = -b$, $2b = 1$
$c = -d$, $2d = 2$

$\Rightarrow$ $a = -\frac{1}{2}$, $b = \frac{1}{2}$, $c = -1$, $d = 1$

$\begin{pmatrix} -\frac{1}{2} & \frac{1}{2} \\ -1 & 1 \end{pmatrix}$ determinant $-\frac{1}{2} - -\frac{1}{2} = 0$

b) $\begin{pmatrix} -\frac{1}{2} & \frac{1}{2} \\ -1 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -\frac{1}{2}x + \frac{1}{2}y \\ -x + y \end{pmatrix}$

so $x' = \frac{1}{2}(y-x)$, $y' = y - x$

so $y' = 2x'$ so all points mapped to $y = 2x$

c) Take line $y = mx + c$

$$\begin{pmatrix} -\frac{1}{2} & \frac{1}{2} \\ -1 & 1 \end{pmatrix} \begin{pmatrix} x \\ mx + c \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

needs to be true for all x

$$\Rightarrow \quad -x + mx + c = 0 \quad \Rightarrow (m-1)x + c = 0$$

so $m = 1$ and $c = 0$ so $y = x$ maps to $(0,0)$

Q11. Let $A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$ $B = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}$

a) $$AB = \begin{pmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} \end{pmatrix}$$

$$\det(AB) = (a_{11}b_{11} + a_{12}b_{21})(a_{21}b_{12} + a_{22}b_{22}) - (a_{11}b_{12} + a_{12}b_{22})(a_{21}b_{11} + a_{22}b_{21})$$

$$= (a_{11}a_{22} - a_{12}a_{21})(b_{11}b_{22} - b_{12}b_{21})$$

$$= \det(A)\det(B)$$

so True and proven

b) $$A+B=\begin{pmatrix} a_{11}+b_{11} & a_{12}+b_{12} \\ a_{21}+b_{21} & a_{22}+b_{22} \end{pmatrix}$$

$$\det(A+B) = (a_{11}+b_{11})(a_{22}+b_{22}) - (a_{12}+b_{12})(a_{21}+b_{21})$$

$$= \left.\begin{aligned}&(a_{11}a_{22}-a_{12}a_{21}) \\ +&(b_{11}b_{22}-b_{12}b_{21})\end{aligned}\right\} \det(A)+\det(B)$$

$$+\ a_{11}b_{22}+b_{11}a_{22}-a_{12}b_{21}-b_{12}a_{21}$$

$$\neq \det(A)+\det(B)$$

So False

Also a simple counter example

$$A=\begin{pmatrix}1&1\\1&1\end{pmatrix} \quad B=\begin{pmatrix}1&2\\2&1\end{pmatrix}$$

Then $\det(A+B)=\det\begin{pmatrix}2&3\\3&2\end{pmatrix}=-5 \neq 0+-3$

$\det(A)=0 \quad \det(B)=-3$